

1. Brockwell and Davis 3.1, 3.11, 4.8, 4.13. For 3.1, you are welcome to check for causality/invertibility with the assistance of a computer in the sense of using a computer program to compute, e.g., roots of a polynomial. You are *not* welcome to find some package that offers a function where you just dump in coefficients and get back the answer.

2. An ARIMA(p,d,q) (where the I is for *integrated*) process is a time series $\{X_t\}$ where $(I - B)^d X_t$ is an ARMA(p,q) process. A SARIMA(s,p,d,q), where the S is for *seasonal*, process is a time series where $(I - B^s)^d X_t$ is an ARIMA(p,d,q) process. Let's explore simulating and fitting these extensions to data. For this question, you are welcome to find a package that fits (S)AR(I)MA models, but find one that lets you specify the degrees s,p,d,q manually! We'll talk about estimation in detail later in the semester, but I don't want to wait too long before asking you to start connecting the math to data.

1. Write a function in whatever programming language you like to simulate an ARMA(p,q) process in the state-space formulation. Importantly, please design this program to accept custom white noise generators. For a Gaussian white noise, generate a length $n = 1,000$ ARMA(p,q) process with, say, $p, q > 3$ and put it into the estimation function that you find to check your work.
2. Now, using the same coefficients, generate a similar length dataset but with $Z_t \sim t_{2.1}$ (the t-distribution with $\nu = 2.1$ degrees of freedom). Run it through your estimation function again and comment on your results. How would you summarize the effect of very heavy tailed noise on generic estimation packages?
3. Download `nao.csv`, the North Atlantic Oscillation time series from Canvas (and note there is a missing value, which you'll have to decide how to address). Using empirical ACFs and pACFs like I quickly demonstrated in class (both on your data and on differenced data and so on), do some exploratory analysis and propose and fit a SARIMA(s,p,d,q) model. Please report your final model and provide justification for your choices. As a final validation, please simulate from your model and repeat the diagnostics on your simulated data. Discuss your results. In particular, try to propose the most important feature of this data that your SARIMA model is not capturing.
- 4* an ARIMA(p,1,q) process is effectively an ARMA(p,q) process with a "unit root", meaning $\phi(z) = 0$ for some $|z| = 1$. Interpret the implication of this addition to the model in the spectral domain. How is the ARMA spectral density impacted? What are the downsides of such a design? Try and propose an extension of ARMA models in the same spirit, but that addresses your identified downside.

3. Consider a stationary time series $\{Y_t\}$ with mean zero and covariance sequence given by

$$\text{Cov}(Y_t, Y_{t-h}) = K(h) = \frac{2e^{-\theta/2}(-\theta \cos(\pi k) + e^{\theta/2}\theta + 2\pi k \sin(\pi k))}{\theta^2 + (2\pi k)^2}.$$

Let's fit an ARMA process to this and see what is going on.

1. Write code to simulate draws from this process of length $n = 1,000$ directly using the Cholesky factor of the covariance matrix. You are welcome to use built-in routines for the Cholesky factorization itself, but please otherwise write this code yourself.
2. Pick orders p and q (again, I suggest trying different combinations for max fun!) and fit an ARMA(p,q) model to that simulation. Plot the implied ARMA ACF against the true ACF and comment on the patterns. You are again welcome to use a package for both fitting and obtaining the ARMA ACF.
- 3*. Show that this $K(h)$ corresponds to a spectral density given by $S(\omega) = e^{-\theta|\omega|}$. Plot this true spectral density, and in the same plot also add the ARMA model-implied SDF using your fitted parameters. Comment on the quality of the fit and any specific issues you see.
- 4*. Google around and choose an algorithm with a convenient package for fitting rational functions (you might look at the *Remez* or *AAA* algorithms, for example). Fit a rational function of the same order as your chosen ARMA process to directly to the true spectral density you verified in the last part, trying carefully to match the way that ARMA SDFs are parameterized. Compare your results with the fitted ARMA coefficients you obtained in 2 and discuss potential differences you observe.